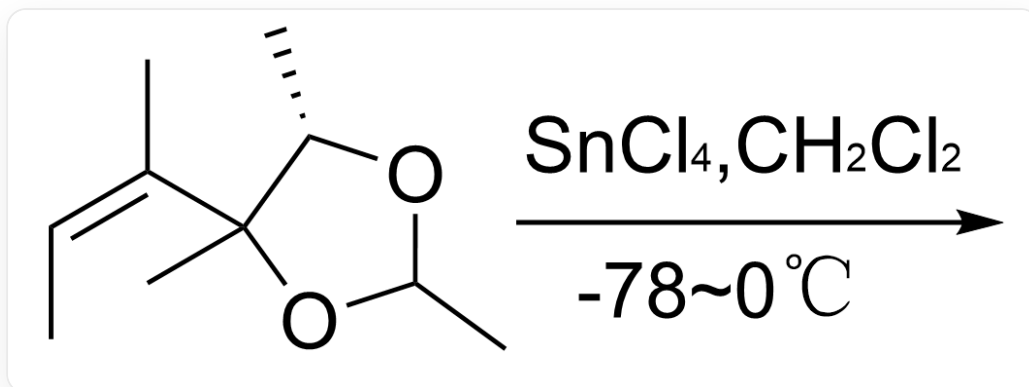


Question

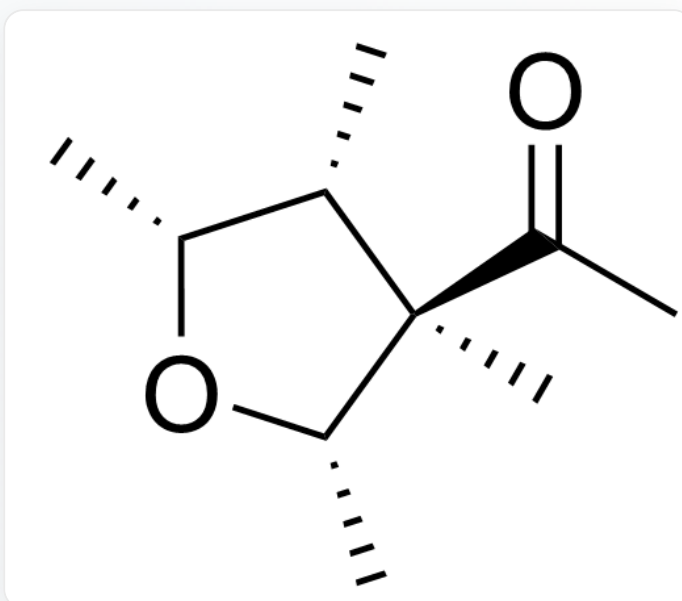
The substrate of the following organic reaction undergoes a transformation under Lewis acid catalysis, selectively yielding another five-membered ring product:



C/C=C(C1([C@@H](OC(O1)C)C)C)/C, the substrate reacts under $-78^\circ\text{C} \sim 0^\circ\text{C}$ conditions in dichloromethane with SnCl_4 as the catalyst

The options provide several possible product structures along with their corresponding absolute configurations. Select the correct one.

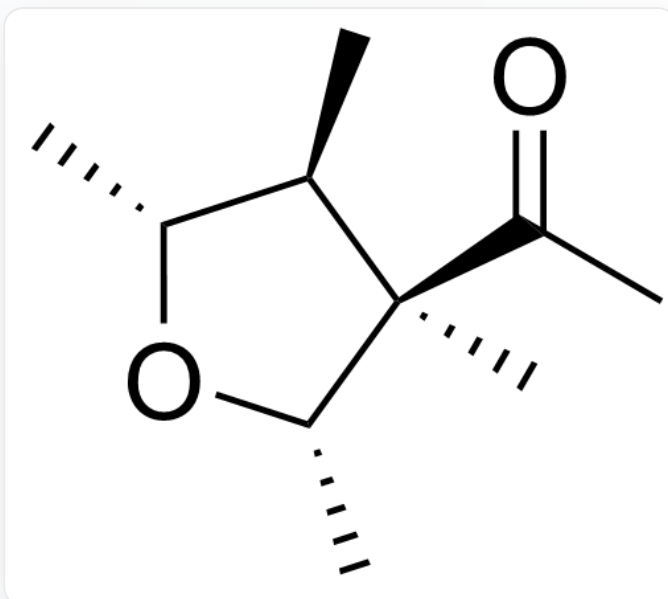
A.



C[C@H]1[C@H](O[C@H]([C@@]1(C(C)=O)C)C)C

Carbon at position 1 on the five-membered ring is S, carbon at position 2 is R, carbon at position 3 is R, and carbon at position 4 is R.

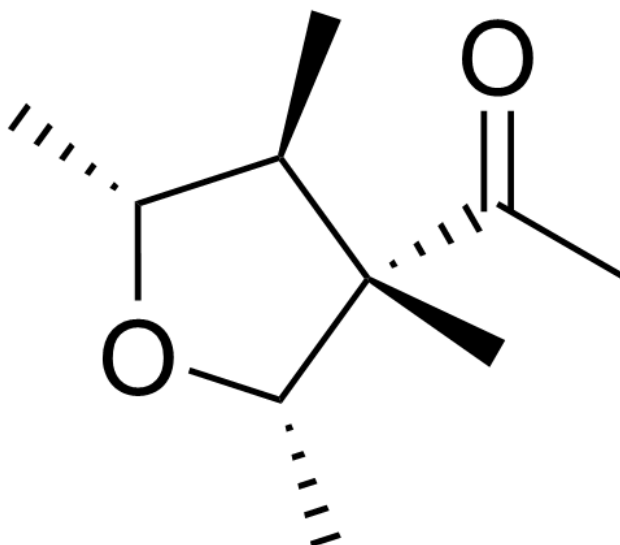
B.



C[C@@H]1[C@H](O[C@H]([C@@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is S, the carbon at position 2 is R, the carbon at position 3 is S, and the carbon at position 4 is R

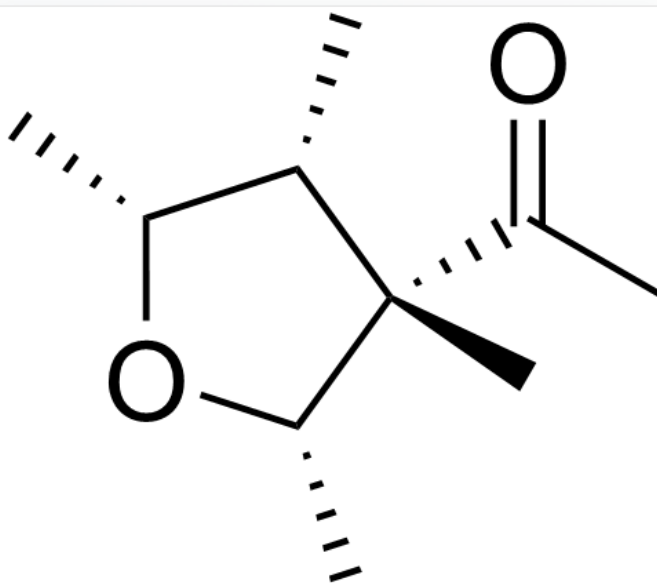
C.



C[C@@H]1[C@H](O[C@H]([C@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is S, the carbon at position 2 is S, the carbon at position 3 is S, and the carbon at position 4 is R.

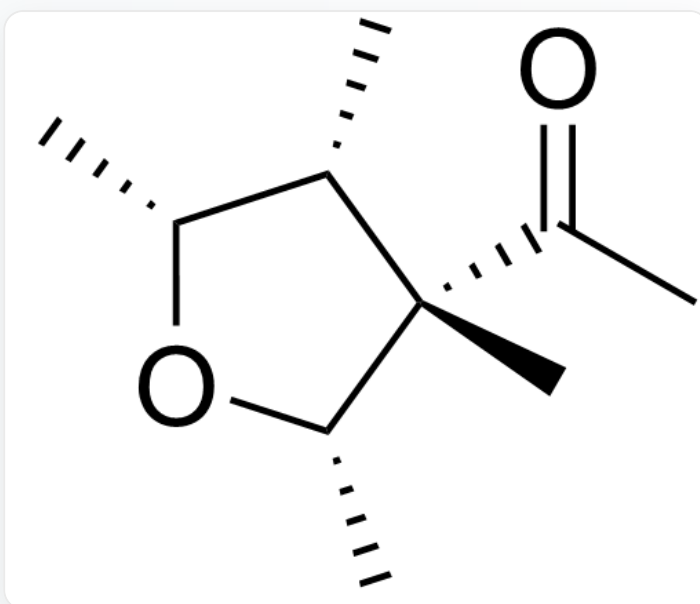
D.



C[C@H]1[C@H](O[C@H]([C@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is S, the carbon at position 2 is S, the carbon at position 3 is R, and the carbon at position 4 is R.

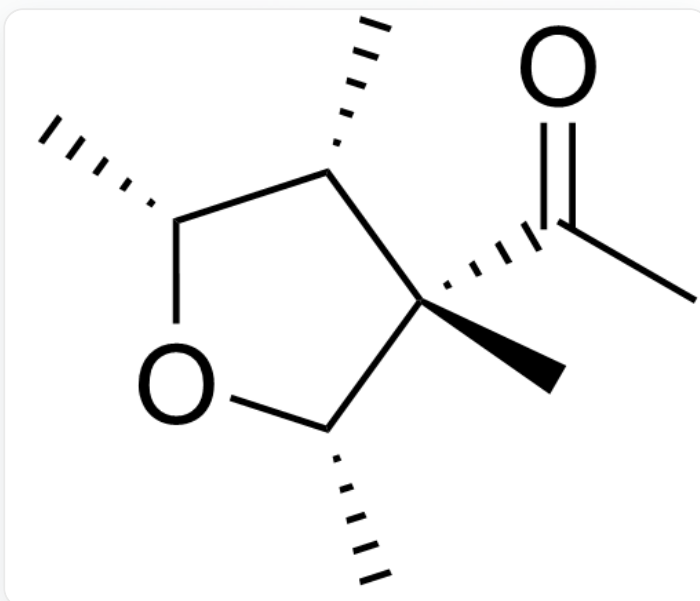
E.



C[C@H]1[C@H](O[C@H]([C@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is R, the carbon at position 2 is S, the carbon at position 3 is R, and the carbon at position 4 is R

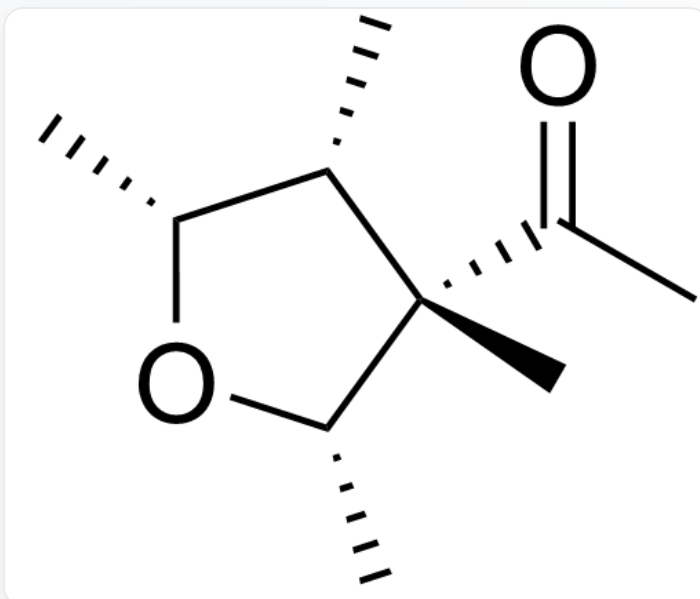
F.



C[C@H]1[C@H](O[C@H]([C@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is S, the carbon at position 2 is R, the carbon at position 3 is R, and the carbon at position 4 is R.

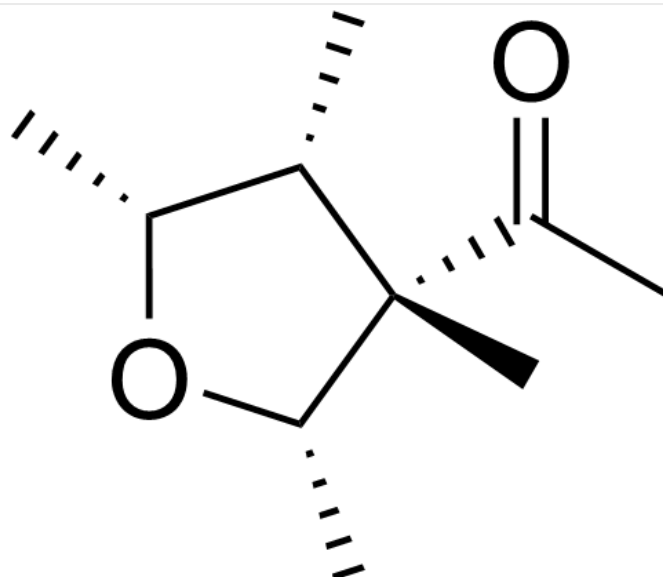
G.



C[C@H]1[C@H](O[C@H]1([C@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is S, the carbon at position 2 is S, the carbon at position 3 is S, and the carbon at position 4 is R.

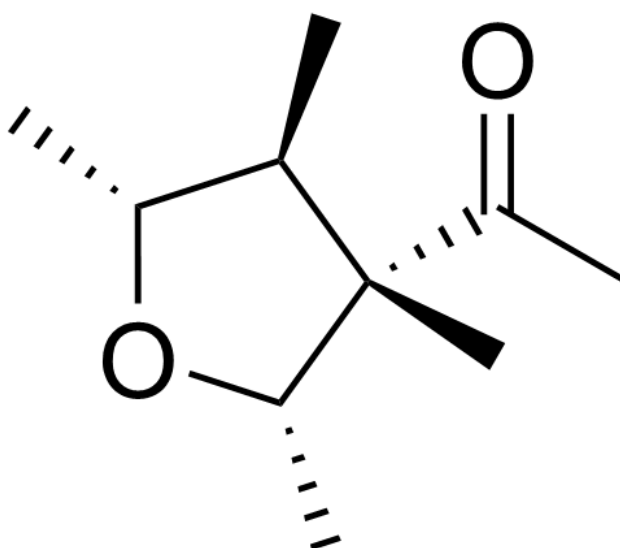
H.



C[C@H]1[C@H](O[C@H]([C@]1(C(C)=O)C)C)C

The carbon at position 1 on the five-membered ring is S, the carbon at position 2 is S, the carbon at position 3 is R, and the carbon at position 4 is S.

I.



C[C@@H]1[C@H](O[C@H]([C@]1(C(C)=O)C)C)C

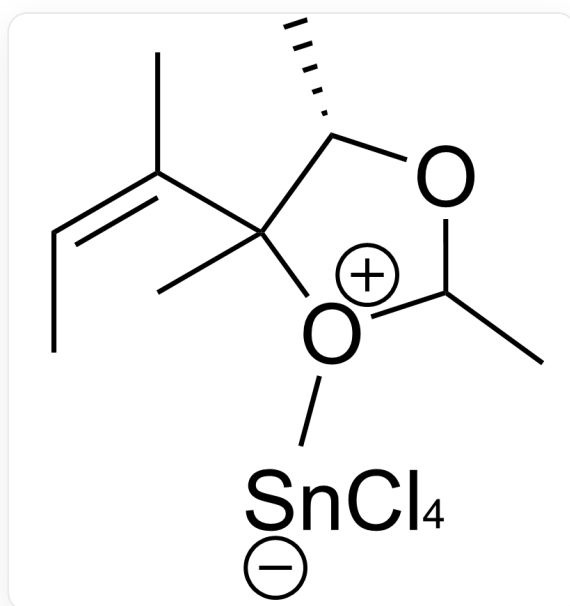
The carbon at position 1 on the five-membered ring is R, the carbon at position 2 is S, the carbon at position 3 is S, and the carbon at position 4 is R.

Answer

Correct Answer: **D**

Detailed Explanation

The five-membered ring of the substrate undergoes ring-opening under Lewis acid catalysis



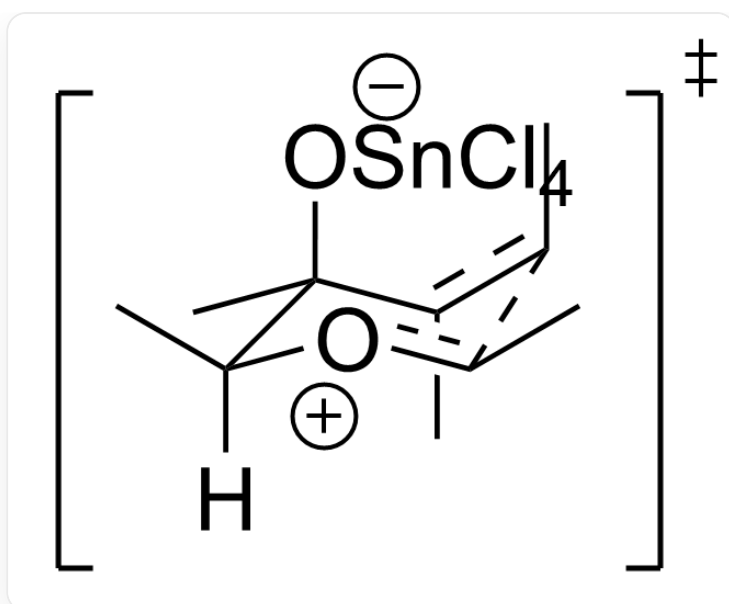
C/C=C(C1([C@@H](OC([O+](C)C)C)C)/C

CHECKPOINT

1 PTS

The five-membered ring of the substrate undergoes ring-opening under Lewis acid catalysis

Through a chair-like transition state, a six-membered ring intermediate is formed via the Prins cyclization process



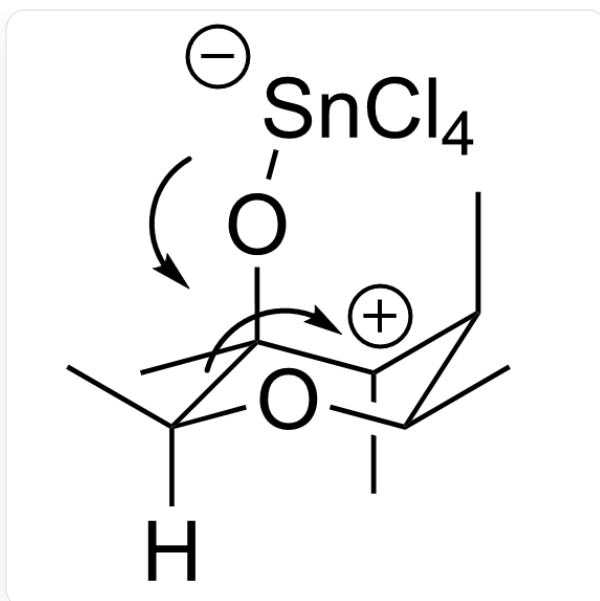
Cc([C@](C)(O[Sn-](Cl)(Cl)(Cl)Cl)[C@@]([H])([O+]c1C)C)c1C

CHECKPOINT

1 PTS

Through a chair-like transition state, a six-membered ring intermediate is formed via the Prins cyclization process

Finally, a five-membered ring is re-formed through a one-step pinacol rearrangement



C[C+](C)[C@H]1C[C@](C)(O[Sn-](Cl)(Cl)Cl)[C@@]([H])(O[C@@H]1C)C

CHECKPOINT

1 PTS

Finally, a five-membered ring is re-formed through a one-step pinacol rearrangement